Validation of the Exact Truncated Zonal Averaged GEqOE Model

This note validates the exact symbolic truncated zonal averaged GEqOE model for the mixed zonal set $\{J_2, J_3, J_4, J_5\}$ against numerical computation from the full GEqOE zonal dynamics.

The reference osculating orbit is

$$a = 16000 \text{ km}, \qquad e = 0.35, \qquad i = 50.0^\circ, \qquad \Omega_0 = 25.0^\circ, \qquad \omega_0 = 40.0^\circ, \qquad M_0 = 30.0^\circ.$$

Two checks were performed:

1. pointwise parity between the exact symbolic averaged RHS and the numerical one-revolution average of the full zonal RHS;
2. long-arc comparison between the propagated exact averaged slow model and per-orbit means extracted from the full zonal GEqOE integration.

Pointwise averaged RHS parity

Component	Earth coefficients	$0.1\times$ coefficients
$\dot{\bar{g}}$	6.815e-14	5.212e-14
$\dot{\bar{Q}}$	2.092e-03	2.151e-04
$\dot{\bar{\Psi}}$	1.499e-04	1.499e-05
$\dot{\bar{\Omega}}$	3.802e-05	3.802e-06

The nonzero residual for $\dot{\bar{Q}}, \dot{\bar{\Psi}}, \dot{\bar{\Omega}}$ shrinks essentially linearly with the zonal scale factor. This is the expected signature of a first-order averaged model compared against the full nonlinear numerical average: the remaining mismatch is higher-order in the zonal strength, not evidence of a wrong first-order closure.

Long-arc slow-flow validation

The exact averaged slow model was initialized from the numerical mean state of the first orbit and then propagated for 30 orbital periods. The resulting mismatch against the subsequent numerical orbit means is:

Quantity	metric
p_1 relative RMS	2.109e-06
p_2 relative RMS	6.308e-06
q_1 relative RMS	6.379e-06
q_2 relative RMS	1.640e-06
Ψ RMS phase error [rad]	1.628e-06
Ω RMS phase error [rad]	3.072e-06

This validates the *slow* exact truncated zonal averaged model. No general zonal short-period reconstruction is used here, so the comparison is deliberately made against orbit means rather than against the full osculating state.

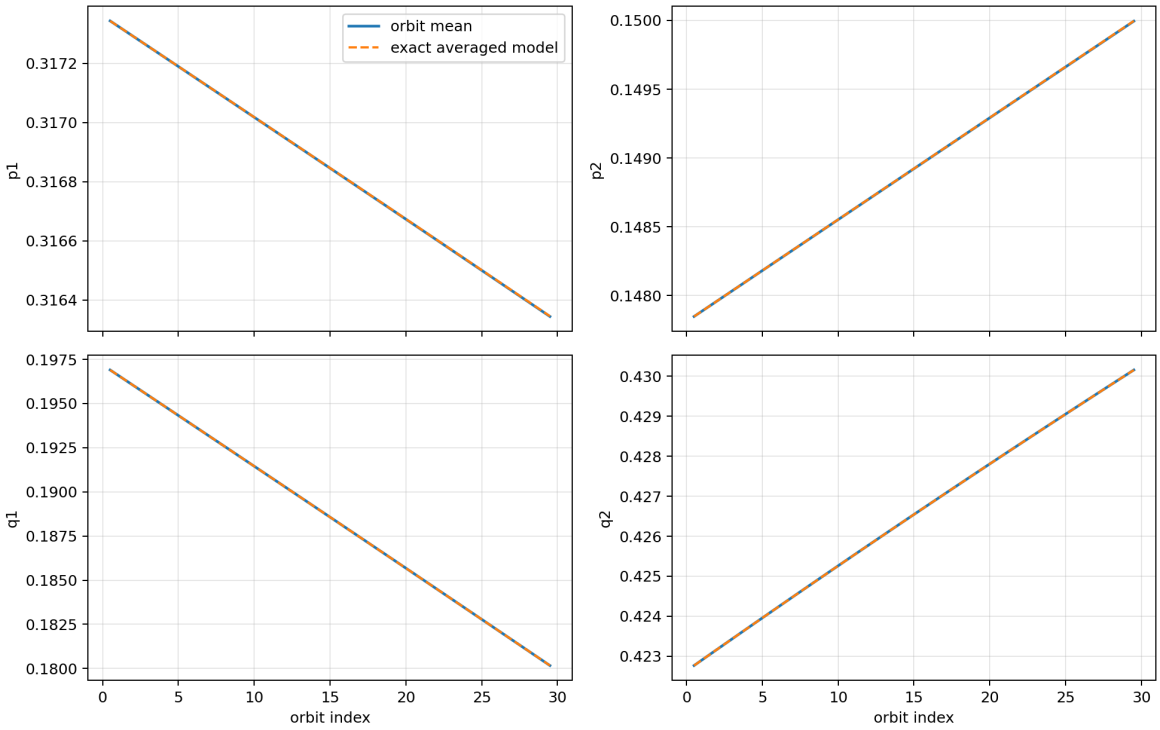

Figure 1: Per-orbit numerical means versus the exact truncated averaged model.

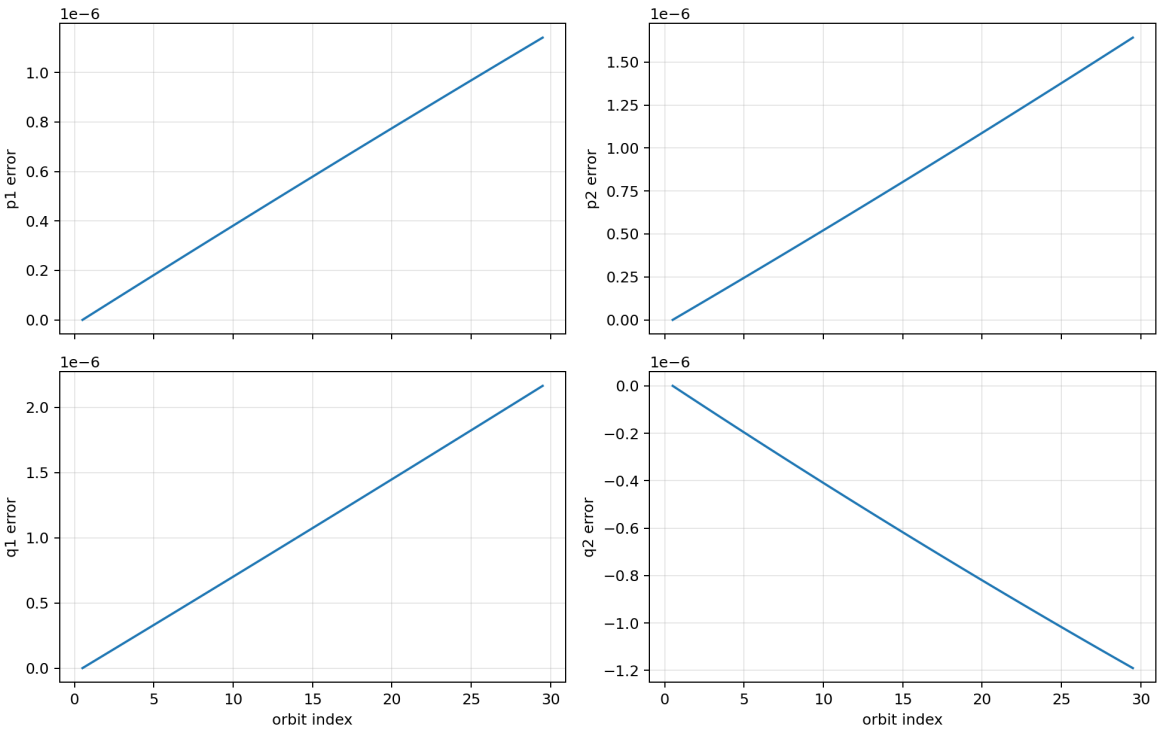

Figure 2: Component-wise slow-state errors of the exact truncated averaged model.